FailWhere: Dataset Documentation and Intended Uses

Supplementary material

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1. Overview

FailWhere is a benchmark for testing whether a vision-language model (VLM) can verbally identify which spatial region of an image caused its own incorrect answer. The release contains: (a) a synthetic grid-image benchmark, (b) a real-image extension built on the Visual Spatial Reasoning (VSR) dataset, (c) counterfactual-masking ground truth (CF-GT) for 142 model failures, and (d) the complete model response logs underlying every number in the paper.

2. Motivation

The benchmark was created to isolate *verbal failure self-localization*: given an answer the model got wrong, can it name the responsible region using only its verbal output? Prior attribution work either requires model internals or only evaluates correct answers. The benchmark is intended for evaluating introspection, faithfulness, and spatial reasoning of VLMs. It is not intended for model training, and it is not a measure of general VQA ability.

3. Composition

3.1 Synthetic split (`code/data/qa_samples_500.json`)

500 samples. Each image is a 300×300 px synthetic grid divided into nine labeled cells (top-left through bottom-right); each cell holds one shape×color object. Two question types: *adjacent_distractor* and *relative_position*. Ground truth is the *answer-cell* (cell containing the correct answer), known by construction.

3.2 VSR split (real images)

The first 500 items of `cambridgeltl/vsr_random`, split `test` (Liu et al. 2023): COCO train2017 photographs paired with binary True/False spatial captions. Model outputs and region tags are released; the underlying images are redistributed only by reference to the VSR/COCO sources (Section 6).

3.3 Counterfactual-masking ground truth (CF-GT)

For each of 142 model failures on the VSR split, each of the nine grid cells was masked with solid gray $(128, 128, 128)$ and the model re-run. A cell whose masking flips the answer to correct is CF-GT; outcomes are labeled `clean` (exactly one cell, n=22), `multiple` (n=47), or `none_found` (n=73).

3.4 Model response logs

Six result JSONs under `code/experiments/` record, per item, the question, model answer, all three self-localization prompt responses (neutral, stateless, multi-turn), the parsed REGION: tag, CF-GT, and per-cell masking answers. A 141-item Sonnet control file (`real_vsr_v3_t0_sonnet_explain.json`) records a second model's attributions of Haiku's failures.

4. Collection and annotation

No human annotators were used. Synthetic items were generated programmatically (`code/generate_synthetic.py`); ground truth is therefore exact by construction. VSR items are unchanged from the source dataset. CF-GT labels were produced mechanically by the masking procedure, not by human judgment. Model generations were produced via the Anthropic API: `claude-haiku-4-5-20251022` / `global.anthropic.claude-haiku-4-5-20251001-v1:0` (temperature 0.0) and `claude-sonnet-5-5` / `global.anthropic.claude-sonnet-5-5` (default temperature; the Bedrock endpoint rejects the parameter for this model). The COCO evidence-cell analysis joins VSR captions to COCO `instances_train2017` bounding boxes via a documented noun-matching heuristic (93.4% join rate; a known limitation).

5. Recommended uses and misuses

Recommended: evaluating verbal self-attribution, faithfulness, and region-grounding of VLMs; counterfactual evaluation methodology research. **Not recommended:** training; measuring general VQA accuracy; drawing conclusions about models' internal (non-verbal) representations, which this benchmark does not access.

6. Distribution, rights, and licensing

Repository: `https://github.com/subodhwandile/vlm-failure-bench`, released under the MIT license. The VSR dataset is a public research dataset built on COCO (which carries its own per-image licenses); this release references those images by the source URLs rather than redistributing them, and users must comply with the COCO terms of use. Model-generated outputs were produced through a paid API and are released by the author. The author bears all responsibility in case of any violation of rights, and confirms the licenses of all released artifacts.

7. Maintenance plan

The benchmark is hosted on GitHub and maintained by the author via the repository issue tracker. Versioned result files are committed with the code; any future data corrections will be issued as tagged releases and documented in the repository changelog. No fixed update cadence is planned; the dataset is a frozen experimental artifact corresponding to the paper's reported results.